MINI PROJECT REPORT

AN INTELLIGENT PERSONALIZED ACADEMIC PLANNING AND PERFORMANCE PREDICTION SYSTEM FOR STUDENTS

A Web-Based AIML Mini Project

Developed Using Base44 AI Application Builder

DECLARATION

We hereby declare that the project entitled **"An Intelligent Personalized Academic Planning and Performance Prediction System for Students"** is our original mini-project work carried out under the guidance of **[Guide Name]**.

The application was designed and deployed using the Base44 AI application-building platform. The work presented in this report has been prepared for academic purposes and has not been submitted elsewhere for the award of any other degree or certification.

ACKNOWLEDGEMENT

We express our sincere gratitude to our project guide, Head of the Department, faculty members, and our institution for their guidance and support throughout this mini project.

We also thank the Base44 platform for providing an AI-assisted environment for designing, developing, testing, and deploying the web application.

Finally, we thank our classmates, friends, and family for their valuable feedback and encouragement.

ABSTRACT

Students often struggle to convert academic goals, subjects, available study time, attendance, assessment marks, and previous performance into a practical study plan. Generic timetables do not consider the individual strengths and weaknesses of a learner and therefore may not provide timely support to students who are at academic risk.

This project proposes an **Intelligent Personalized Academic Planning and Performance Prediction System for Students**, a web-based application that organizes student academic information and uses intelligent, data-driven logic to generate personalized study recommendations and performance insights.

The system is designed around student profiles, subjects, study goals, academic performance, planned activities, and prediction-oriented indicators. Based on the available academic data, the application can help a student identify priority subjects, plan study activities, monitor progress, and understand possible future performance.

The project is implemented using **Base44**, an AI-powered application development platform capable of generating application interfaces, data structures, backend logic, authentication, and hosting from natural-language requirements.

The resulting application demonstrates how AI-assisted/no-code development can be used to create a practical educational technology solution within a short development cycle.

Keywords: Artificial Intelligence, Machine Learning, Personalized Learning, Academic Planning, Performance Prediction, Student Analytics, Web Application, Base44 AI.

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1. INTRODUCTION

Academic success depends not only on the amount of study but also on how effectively a student allocates time among subjects and responds to performance trends.

Students commonly maintain marks, attendance, assignments, and study schedules in different places, making it difficult to obtain a unified view of their academic progress. A personalized planning system can address this problem by combining academic records with planning and analytical features.

The proposed **Intelligent Personalized Academic Planning and Performance Prediction System for Students** provides a centralized web application in which a student can maintain academic information, define goals, plan study activities, monitor progress, and obtain performance-oriented insights.

The project combines the principles of Artificial Intelligence and Machine Learning with a practical web-based student-support application.

The application is developed using **Base44 AI**, which enables rapid development of web applications through AI-assisted natural-language instructions.

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2. PROBLEM STATEMENT

Students face difficulties in managing multiple subjects, assignments, examinations, projects, and study schedules.

Traditional academic planning methods are usually static and require students to manually analyze their performance. They may not provide personalized recommendations based on individual strengths, weaknesses, study habits, and academic goals.

Therefore, there is a need for an intelligent web-based system that can use student academic and planning data to generate personalized study recommendations and performance-oriented predictions.

The proposed system aims to help students:

- Prioritize subjects.
- Manage study time.
- Monitor academic progress.
- Identify weak areas.
- Understand performance trends.
- Take corrective action before performance declines.

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3. OBJECTIVES

The main objectives of the project are:

1. To create a centralized platform for maintaining student academic information.
2. To analyze marks, study patterns, goals, and progress indicators.
3. To generate personalized academic plans based on student-specific needs.
4. To identify subjects or activities that require higher priority.
5. To provide performance-oriented predictions or risk indicators.
6. To present academic information through simple dashboards and visual summaries.
7. To enable continuous monitoring of academic progress.
8. To help students improve their time-management skills.
9. To provide personalized academic recommendations.
10. To demonstrate AI-assisted web application development using Base44 AI.

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4. SCOPE OF THE PROJECT

The scope of the proposed system includes:

- Student profile management.
- Academic information management.
- Subject management.
- Subject-wise performance tracking.
- Study goal management.
- Study task management.
- Personalized study planning.
- Subject prioritization.
- Progress monitoring.
- Performance trend analysis.
- Prediction-oriented insights.
- Personalized recommendations.
- Dashboard-based visualization.
- Web-based access.

The system can be further extended to support teachers, mentors, and educational institutions.

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5. EXISTING SYSTEM

Traditional academic planning is generally performed using:

- Manual timetables.
- Notebooks.

- Spreadsheets.
- Generic task-management applications.
- Basic learning portals.

These approaches require students to manually manage and analyze their academic information.

5.1 Limitations of Existing System

1. Manual Planning

Students need to manually create and modify their study schedules.

2. Static Timetables

Traditional timetables may not change according to improvements or declines in academic performance.

3. Limited Personalization

Generic planning systems usually do not consider individual strengths and weaknesses.

4. Manual Performance Analysis

Students need to calculate and interpret their academic performance themselves.

5. Lack of Prediction

Traditional planning tools generally do not provide future performance or academic-risk indicators.

6. Lack of Centralization

Academic information, study plans, marks, and goals may be stored in different places.

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6. PROPOSED SYSTEM

The proposed **Intelligent Personalized Academic Planning and Performance Prediction System for Students** combines academic data management, personalized planning, progress monitoring, and performance-oriented analysis in one web application.

The system accepts relevant student information, organizes it into structured records, analyzes available indicators, and presents personalized academic insights.

6.1 Major Features

- Student academic dashboard.

- Student profile management.
- Subject management.
- Academic performance tracking.
- Personalized study planning.
- Study task management.
- Progress monitoring.
- Subject priority analysis.
- Performance prediction/risk indication.
- Personalized recommendations.
- Visual academic summaries.

The system is intended to support student decision-making rather than replace teachers or official institutional evaluation.

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7. SYSTEM REQUIREMENTS

7.1 Hardware Requirements

- Computer or laptop.
- Smartphone/tablet with modern browser.
- Minimum 4 GB RAM recommended.
- Stable internet connection.

7.2 Software Requirements

- Windows/Linux/macOS/Android/iOS.
- Google Chrome, Microsoft Edge, Mozilla Firefox, or equivalent browser.
- Base44 AI application builder.
- Internet connection.

The primary Base44 development workflow does not require a conventional local programming environment.

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8. SYSTEM ARCHITECTURE AND METHODOLOGY

The system follows a **data-to-insight workflow**.

Overall Architecture

Student

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Web Interface

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Academic Data Management

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Application Logic

↓

Intelligent Analysis

↓

Performance Prediction & Recommendations

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Dashboard

8.1 Presentation Layer

This layer provides:

- Web pages.
- Forms.
- Dashboard.
- Navigation.
- Charts and summaries.
- Student interaction screens.

8.2 Application Logic Layer

This layer handles:

- Input validation.
- Calculations.
- Subject prioritization.
- Study-plan logic.
- Progress calculations.
- Recommendation logic.

8.3 Data Layer

This layer stores:

- Student information.
- Subjects.
- Performance records.
- Study plans.
- Goals.
- Tasks.
- Progress.
- Prediction information.

8.4 Intelligence Layer

This layer is responsible for:

- Performance analysis.
- Trend identification.
- Subject prioritization.
- Risk/score estimation.
- Personalized recommendations.

8.5 Deployment Layer

The application is deployed as a web application using Base44.

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9. FUNCTIONAL MODULES

9.1 Student Profile Module

This module stores student information required for personalization.

Possible information includes:

- Student name.
- Branch.
- Semester.
- Academic preferences.
- Study goals.

9.2 Subject Management Module

This module allows students to manage their subjects and related academic information.

9.3 Academic Performance Module

This module records:

- Examination marks.
- Assessment scores.
- Subject-wise performance.
- Performance trends.

9.4 Study Planning Module

This module allows students to create study activities and organize their study time.

Students can specify:

- Subject.
- Date.
- Study duration.
- Task.
- Priority.
- Completion status.

9.5 Progress Monitoring Module

This module tracks the completion of study activities and academic progress.

9.6 Performance Prediction Module

This module uses available academic indicators to estimate future performance or identify possible academic risk.

9.7 Recommendation Module

This module converts analysis into useful academic suggestions.

Examples include:

- Focus more on weak subjects.
- Increase study time.
- Complete pending tasks.
- Revise before examinations.
- Prioritize important subjects.

9.8 Dashboard Module

The dashboard provides a centralized view of:

- Academic performance.
- Study progress.
- Priority subjects.
- Goals.
- Recommendations.
- Prediction-oriented insights.

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10. AIML / PREDICTION METHODOLOGY

The intelligence component of the project is based on analyzing measurable academic indicators.

10.1 Possible Input Features

The system can consider available information such as:

- Previous examination marks.
- Recent assessment scores.
- Subject-wise performance.
- Planned study time.
- Study-task completion.
- Academic goals.
- Progress percentage.
- Attendance, if available.

10.2 Prediction Concept

The system can map academic indicators to:

- Predicted score.
- Performance category.
- Academic risk level.
- Subject priority.

For example, consistently low marks combined with low study-plan completion may indicate that a student needs additional attention.

Similarly, improving marks and consistent study progress may indicate a positive performance trend.

10.3 Possible Machine Learning Algorithms

Prediction Task	Possible Algorithm
Numerical score prediction	Linear Regression
Performance classification	Logistic Regression
Risk prediction	Decision Tree
Advanced prediction	Random Forest
Student grouping	K-Means Clustering
Advanced prediction	Gradient Boosting

Note: These algorithms should be reported as implemented algorithms only if they are actually trained and evaluated in the project. If the current application uses intelligent rules or AI-generated recommendations, it should be described as an intelligent/rule-based prediction prototype.

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11. BASE44 AI DEVELOPMENT METHODOLOGY

The project is developed using **Base44 AI**, an AI-assisted application development platform.

11.1 Development Process

Step 1: Requirement Analysis

The academic planning and performance prediction requirements were identified.

Step 2: Natural-Language Prompting

The application requirements were provided to Base44 AI using natural-language instructions.

Step 3: Application Generation

Base44 AI was used to generate the initial application structure and interface.

Step 4: UI Customization

The generated interface was refined according to the project requirements.

Step 5: Data Modelling

The required academic data entities were defined.

Step 6: Logic Development

Application logic was refined for:

- Performance analysis.
- Study planning.
- Progress monitoring.
- Recommendations.
- Prediction-related functionality.

Step 7: Testing

The application was tested using different user inputs and workflows.

Step 8: Deployment

The application was deployed as a web application.

Project Application

Study Smart – Savvy Path

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12. DATABASE AND DATA MODEL

The proposed data model consists of the following entities:

Entity	Important Attributes
Student	Student ID, Name, Email, Branch, Semester
Subject	Subject ID, Subject Name, Credits, Target Score
Performance	Performance ID, Student ID, Subject ID, Assessment, Score, Date
Study Plan	Plan ID, Student ID, Subject ID, Date, Duration, Priority, Status
Goal	Goal ID, Student ID, Description, Target Date, Status
Progress	Progress ID, Student ID, Task ID, Completion Percentage
Prediction	Prediction ID, Student ID, Predicted Score/Risk, Date

The exact entity names and fields should match the final Base44 application configuration.

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13. USER INTERFACE AND WORKFLOW

13.1 Main Screens

Landing Page

Introduces the application and provides access to the student workspace.

Student Dashboard

Displays academic summaries, progress, priorities, and insights.

Student Profile

Stores student-specific information.

Subject and Performance Page

Displays subjects and academic performance information.

Study Planner

Allows students to create and manage study activities.

Prediction/Insights Page

Displays performance estimates, trends, risk indicators, and recommendations.

Progress Page

Displays study completion and academic progress.

13.2 User Workflow

1. Open the application.
2. Register or login if authentication is enabled.
3. Complete the student profile.
4. Add academic subjects.
5. Enter performance information.
6. Set academic goals.
7. Create study plans.
8. Review priorities and recommendations.
9. Follow the study plan.
10. Update progress.
11. Review updated performance insights.

14. TESTING AND VALIDATION

Testing is performed to verify the correctness and usability of the application.

Test ID	Test Case	Expected Result
TC01	Open application	Application loads successfully
TC02	Enter student details	Data is accepted
TC03	Enter invalid data	Validation message is displayed
TC04	Add subject	Subject is added successfully
TC05	Add performance	Performance information is stored
TC06	Create study plan	Study task appears in planner
TC07	Update progress	Progress is updated
TC08	Generate insight	Academic insight is displayed
TC09	Navigate between pages	Correct page opens
TC10	Test different screen sizes	Application remains usable

14.1 Validation Areas

- Functional correctness.
- Data validation.
- Data persistence.
- Recommendation consistency.
- Dashboard usability.
- Browser compatibility.
- Mobile responsiveness.
- Error handling.

15. RESULTS AND DISCUSSION

The developed system demonstrates the feasibility of creating an academic-support application using an AI-assisted development platform.

The system brings academic information and study-planning activities into a single web application.

The project helps students to:

- Organize academic information.
- Plan study activities.
- Identify priority subjects.
- Monitor progress.
- Understand performance trends.
- Receive personalized academic insights.

The application demonstrates the practical use of AI-assisted application development in the field of educational technology.

For a complete Machine Learning evaluation, a larger academic dataset can be used with a proper training and testing process. Metrics such as **MAE, RMSE, Accuracy, Precision, Recall, and F1-Score** can be included depending on the prediction task.

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16. ADVANTAGES

1. Provides personalized academic planning.
2. Helps students manage study time.
3. Centralizes academic information.
4. Helps identify weak subjects.
5. Supports progress monitoring.
6. Provides performance-oriented insights.
7. Provides a simple web interface.
8. Can be accessed through a browser.
9. Enables rapid development using Base44 AI.
10. Can be extended with advanced Machine Learning models.

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17. LIMITATIONS

1. Prediction quality depends on the quality of student data.
2. Incorrect or incomplete data can affect recommendations.
3. A small dataset may not provide reliable ML predictions.
4. Self-reported study information may contain inaccuracies.
5. Internet connectivity is required.
6. Predictions should be considered guidance rather than official academic results.
7. Advanced ML training and evaluation are required for production-level prediction accuracy.

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18. FUTURE ENHANCEMENTS

18.1 Advanced Machine Learning

A larger academic dataset can be used to train and evaluate advanced Machine Learning models.

18.2 AI Study Assistant

An AI assistant can help students with:

- Subject doubts.
- Revision planning.
- Exam preparation.
- Study-resource recommendations.

18.3 Attendance Integration

Attendance information can be integrated to improve academic analysis.

18.4 Adaptive Timetable

The system can automatically generate timetables based on:

- Subject performance.
- Available study time.
- Upcoming examinations.
- Subject difficulty.
- Student progress.

18.5 Notifications

Notifications can be provided for:

- Study tasks.
- Assignments.
- Examinations.
- Pending activities.

18.6 What-If Analysis

Students could estimate how changes in study time may affect expected performance.

18.7 Faculty Dashboard

A faculty dashboard could provide mentors with privacy-aware academic insights.

18.8 Multilingual Support

The system can be extended to support multiple languages for better accessibility.

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19. CONCLUSION

The **Intelligent Personalized Academic Planning and Performance Prediction System for Students** addresses a practical problem faced by students in managing academic activities and monitoring performance.

The system combines academic information management, personalized study planning, progress monitoring, and intelligent performance insights in a single web application.

The project demonstrates how **Artificial Intelligence, Machine Learning concepts, and AI-assisted application development** can be applied to educational technology.

The use of **Base44 AI** helps reduce application development complexity and enables rapid creation and deployment of a functional web application.

The system can be further improved by integrating a properly trained and evaluated Machine Learning model using a sufficiently large and representative academic dataset.

Overall, the project provides a foundation for a personalized academic assistance system that can help students make better study decisions and work towards their academic goals.

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20. REFERENCES

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21. APPENDIX · SCREENSHOTS

The following screenshots can be inserted from the final deployed application:

Figure 1: Home/Landing Page

Figure 2: Student Dashboard

Figure 3: Student Profile / Academic Information

Figure 4: Subject and Performance Entry

Figure 5: Personalized Study Planner

Figure 6: Performance Prediction / Academic Insights

Figure 7: Progress Monitoring Dashboard

Figure 8: Responsive Mobile View

PROJECT DETAILS

Item	Details
Project Title	**An Intelligent Personalized Academic Planning and Performance Prediction System for Students**
Project Type	AIML Mini Project
Development Platform	Base44 AI
Domain	Artificial Intelligence and Machine Learning / Educational Technology
Project Application	Study Smart – Savvy Path
Student Names	[Fill here]
Roll Numbers	[Fill here]
Guide Name	[Fill here]
Department	Artificial Intelligence and Machine Learning
College / University	[Fill here]
Academic Year	2026–2027